

SooHyun Yang

Google Scholar: <https://scholar.google.com/citations?user=zczJPkAAAAJ&hl=en>

Github: <https://github.com/Yang-SooHyun>

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EDUCATION

University of Seoul

Master of Science in Environmental Engineering

GPA: 4.4/4.5

Thesis: Spatiotemporal Responses of Global Lake Algal Blooms to Drought: An Analysis Using a High-Resolution Chlorophyll-*a* Dataset (Tentative)

Advisor: YoonKyung Cha

Seoul, South Korea

Sep. 2025 – Feb. 2027

University of Seoul

Bachelor of Science in Environmental Engineering

GPA: 3.8/4.5

Thesis: Development of Bayesian Network for Probabilistic Mid-to Long-term Forecasting for Harmful Algal Blooms

Seoul, South Korea

Mar. 2020 – Aug. 2025

RESEARCH INTERESTS

Processing satellite observations, implementing robust quality control frameworks, and constructing long-term high-resolution geospatial datasets.

Analyzing long-term variability, identifying trends on the Earth's surface, and modeling spatiotemporal environmental dynamics under climate conditions.

Applying advanced neural network architectures for spatiotemporal data imputation, time-series reconstruction, and predictive environmental modeling.

Leveraging spatial deep learning, semantic segmentation, and object detection for automated feature extraction, land classification, and surface anomaly detection.

RESEARCH EXPERIENCE

University of Seoul

Graduate Research Assistant, Water Environment Management Lab

Seoul, South Korea

Sep. 2025 – Feb. 2027

- Developed a cyanobacteria cell counts forecasting model utilizing deep learning models (Transformer, TFT, TCN, LCA).
- Imputed daily water quality and cyanobacteria cell counts data from mixed-frequency inputs using SAITS (Self-Attention-based Imputation for Time Series).
- Developed a transformer-based hierarchical deep learning model for satellite Chlorophyll-*a* prediction.
- Calibrated and adapted established bio-optical algorithms to optimize Chlorophyll-*a* retrieval accuracy for the unique optical properties of diverse global lakes.
- Constructed GLCD250-MOD, a daily, 250 m Chlorophyll-*a* dataset covering 465,966 inland lakes worldwide from 2000–2024.
- Developed a large-scale pipeline to acquire and process hundreds of terabytes of Terra MODIS imagery, including parallel downloading and polygon-based ROI extraction.
- Analyzed global spatiotemporal trends using GLCD250-MOD and evaluated the response and interaction mechanisms of lake ecosystems with conventional environmental variables under climate change.

Undergraduate Research Assistant, Water Environment Management Lab

Sep. 2023 – Aug. 2025

- Conducted a long-term spatiotemporal trend analysis using the Mann-Kendall test.
- Constructed a Lake Erie Chlorophyll-*a* dataset using open-source data from in-situ and satellites (ENVISAT MERIS, Aqua/Terra MODIS, Sentinel-3A/B OLCI, S-NPP/NOAA-20 VIIRS).
- Projected mid-to long-term algal blooms using Shared Socioeconomic Pathways (SSP).
- Developed a prediction model for cyanobacteria cell counts using Bayesian networks.
- Created a web-specific automatic crawling code.

TEACHING AND MENTORING EXPERIENCE

Engineering Statistics

Teaching Assistant

Mar. 2026 – Jun. 2026

Answered course-related questions, created examples and homework programming codes.

Graduation Thesis Class for Undergraduate Students

Academic Mentor

2026

Guided three undergraduates to successfully complete their bachelor's thesis.

- Thesis 2026: Correlation Analysis between Drought Indices and Algal Bloom Indices for Global Lakes

HONORS & AWARDS

Graduate

Outstanding Oral Presentation Award

Feb. 2026

Graduate Student Research Competency Enhancement Program, Spring

Apr. 2026 – Jul. 2026

Instruction Assistance Scholarship

Mar. 2026 – Jun. 2026

The Combined BS/MS Degree Scholarship (Full tuition)

Sep. 2025 – Feb. 2027

Undergraduate

Outstanding Conference Poster Award

Feb. 2025

Outstanding Graduate Thesis Award

Feb. 2025

Grand Prize, Innovative Talent Cultivation Project for Digital Water Industry

Dec. 2024

Student Future Planning Semester: Research Internship

Jul. 2024 – Aug. 2024

Innovative Talent Cultivation Project for Digital Water Industry

Sep. 2023 – Dec. 2024

Student Future Planning Semester: Global Self-Directed Research

Sep. 2023 – Dec. 2023

PUBLICATIONS

- [1] **Yang, S.**¹, Lee, H.¹, Lee, G., Gu, T., Park, T., Park, J., Shin, J., Kim, T., Cha, Y. (2026). GLCD250-MOD: Global Daily 250 m Chlorophyll-*a* Dataset for Inland Lakes based on MODIS, *Earth System Science Data (In review)*.
- [2] Park, T., Jung, J., Park, J., **Yang, S.**, Moon, J., & Cha, Y. (2026). Structurally Constrained Hierarchical Learning for Improving River Health Assessment using Tabular Transformers. *Water Research (In review)*.
- [3] Kim, T., Lee, H., **Yang, S.**, Lee, G., Shin, J., & Cha, Y. (2025). Long-term Spatiotemporal Variability and Regime Classification of Chlorophyll-*a* Concentrations in Lake Erie Using Satellite Products. *Harmful Algae*, 102896.

CONFERENCES

- [1] **Yang, S.**, Lee, H., Kim, T., & Cha, Y. Long-term Trend Analysis of Algal Blooms in Global Lakes. International Water Association World Water Congress & Exhibition, 2026.
- [2] Lee, H., **Yang, S.**, Kim, T., & Cha, Y. Construction of a Long-term High Spatiotemporal Resolution Chlorophyll-*a* Dataset for Global Lakes. Korean Society on Water Environment – Korean Society of Water & Wastewater Joint Spring Conference, 2026.
- [3] **Yang, S.**, Lee, H., Park, T., Gu, T., Moon, J., Park, S., & Cha, Y. Advancement of Aquatic Ecosystem Model Development and Integration Strategies within a Digital Twin Korea-Water. Korean Society of Limnology Winter Conference, 2026.
- [4] Kim, T., Lee, H., **Yang, S.**, & Cha, Y. Does drought intensify algal blooms in global lakes? American Geophysical Union Winter Conference, 2025.
- [5] Lee, H., **Yang, S.**, Jeong, J., Hwang, G., Moon, J., & Cha, Y. Development of a Machine Learning-Based Geographically Weighted Model for Predicting a Integrated Assessment Index of River Health. Korean Society on Water Environment – Korean Society of Water & Wastewater Joint Fall Conference, 2025.
- [6] **Yang, S.**, Park, T., Park, J., Gu, T., & Cha, Y. Habitat Suitability Assessment and Derivation of Optimal Environmental Conditions in the Geum and Nakdong River Basins Using a Data-Driven HSI Model. Korean Society of Limnology Summer Conference, 2025.

- [7] **Yang, S.**, Gu, T., Park, T., Lee, H., & Cha, Y. Development of Mid-to Long-term Forecasting Model of Harmful Algal Blooms Based on Probabilistic Predictions Using a Bayesian Network. Korean Society of Limnology Winter Conference, 2025.
- [8] Kim, T., Lee, G., Lee, H., **Yang, S.**, Shin, J., & Cha, Y. Characterizing Long-term Spatiotemporal Variability and Regime Heterogeneity of Chlorophyll-a in Lake Erie Using Multiple Satellite-based Products. American Geophysical Union Winter Conference, 2024.
- [9] **Yang, S.**, Lee, H., Kim, T., Shin, J., & Cha, Y. Spatiotemporal long-term pattern analysis and regime classification in Lake Erie using an optimal satellite product. Korean Society of Environmental Engineers Winter Conference, 2024.
- [10] **Yang, S.**, Lee, H., Lee, G., & Cha, Y. Application of Explainable Deep Learning Model for River Eutrophication Assessment. Applied Artificial Intelligence Conference, 2024.
- [11] **Yang, S.**, Jeong, B., & Cha, Y. Analysis of the relationship between the exogenous factor to the Harmful Algae Blooms in the Sejong Weir based on PLS-SEM. Applied Artificial Intelligence Conference, 2023.

PROJECTS

- [1] Green Resource Engineering and Advanced Transformation (GREAT) Center for Waste-to-Wealth, National Research Foundation, Graduate Researcher, Jul. 2026 –Present
- [2] Development of a Mechanism-Based River Water Quality Model Integrated with Artificial Intelligence (II), National Institute of Environmental Research (NIER), Graduate Researcher, Jun. 2026 – Mar. 2027.
- [3] Digital Twin Korea-Water Development Project (5/5): A Study on Optimal Habitat Environmental Conditions for the Application of Ecological Models (II), National Institute of Environmental Research (NIER), Graduate Researcher, Mar. 2026 – Jan. 2027.
- [4] Development of a Mechanism-Based River Water Quality Model Integrated with Artificial Intelligence (I), National Institute of Environmental Research (NIER), Graduate Researcher, May 2025 – Feb. 2026.
- [5] Digital Twin Korea-Water Development Project (4/5): A Study on Optimal Habitat Environmental Conditions for the Application of Ecological Models (I), National Institute of Environmental Research (NIER), Graduate Researcher, Mar. 2025 – Dec. 2025.
- [6] Development and verification of digital twin-based water quality prediction, National Information Society Agency (NIA), Undergraduate Research Assistant, May. 2024 – Dec. 2024.
- [7] Development and Pilot Application of a Digital Twin Aquatic Ecosystem Model (II), National Institute of Environmental Research (NIER), Undergraduate Research Assistant, Mar. 2024 – Dec. 2024.
- [8] Development of a modularized AI-based model for the multiscale analysis of multimodal environmental data, Ministry of Science and ICT (MSIT), Undergraduate Research Assistant, Mar. 2024 – Feb. 2025.
- [9] Water Policy Outlook Based on Integrated Analysis of Water, Environment, and Economy(I), Korea Environment Institute (KEI), Undergraduate Research Assistant, Mar. 2024 – Dec. 2024.

SKILLS

Software

Python, R, ArcGIS, Google Earth Engine, Netica, & Crawling.

Modeling Techniques

Bayesian statistics, Bayesian network, Structural equation modeling, Deep learning, & Machine learning.

Languages

Korean (native), English (advanced), Spanish (elementary), & Mandarin (elementary).

ACTIVITIES & INTERESTS

Falcons, University of Seoul baseball club

Club Member

Sep. 2020 – Present

Played baseball as a second baseman and pitcher.

Other Interests

Baseball, Flute, and Trekking

MILITARY SERVICE

Republic of Korea Army

Jan. 2021 – Jul. 2022

Completed military service as a drill instructor for basic military training.